# Endpoint and Exposure Sensitivity in Fault Assessment of a Fly-Connectome Spiking Model

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***Abstract*—Fault assessment can conflate preserved neural transmission with task tolerance when cohorts, exposure and control laws differ. We quantify these dependencies in a 138,639-neuron *Drosophila* spiking model. A frozen-calibration comparison crosses two laws with healthy, 25%-deletion and 50%-deletion networks on six paired seed clusters, comprising 72 records. Collision-free counts are 4/4/0 of 12 under the original law and 6/3/0 under pre-cue gating. Gated-minus-original net fault loss is +25.0 and +16.7 percentage points at the two doses. The 25% conditional percentile range excludes zero, but only three clusters have nonzero contrasts; a post-hoc sign-flip sensitivity gives $p = 0.25$. All factorial $+30^\circ$ episodes collide. Archive reanalysis removes 12 of 14 additional rejections relative to transmission-only scoring when targeted hubs are excluded. Duration matching changes the median deletion spike ratio from 0.382 to 0.729; six equal-duration hub pairs instead cross twofold at 840–900 ms on calibration scenarios. These conditional findings support reporting cohorts, paired exposure, analysis horizons and task outcomes together, without establishing a resource budget or broad reliability.**

***Index Terms*—neuromorphic computing, connectome, fault assessment, fault injection, Drosophila**

## I. Introduction

Fault tolerance inferred from a neural endpoint can depend on which fault cohorts, observation durations and feedback laws enter the assessment. Preserved command transmission can coexist with elevated activity or failed behavior. The adult *Drosophila* wiring diagram [1], [2] and an executable spiking model [3] provide an anatomically constrained substrate for quantifying this dependence. Biological wiring supplies a structural prior without validating a simplified controller's reliability.

We study software perturbations of a 138,639-neuron leaky integrate-and-fire (LIF) model, driving LC4/LPLC2 visual neurons and reading eight descending neurons (Fig. 1). Five families alter weights, delete weighted connection rows or silence neuronal outputs. These experiments specify neither hardware-resource mapping nor biological resilience.

The contribution is empirical: we quantify how assessment choices change the interpretation of fault effects in this model. Archive reanalysis holds observations fixed while varying cohort and exposure definitions; prospective tests cross two frozen control laws with healthy, 25%-deletion and 50%-deletion networks. Paired task loss separates fault effects from baseline controller success. Duration matching, complete trajectories and endpoint separation are established evaluation practices, not new algorithms.[1]

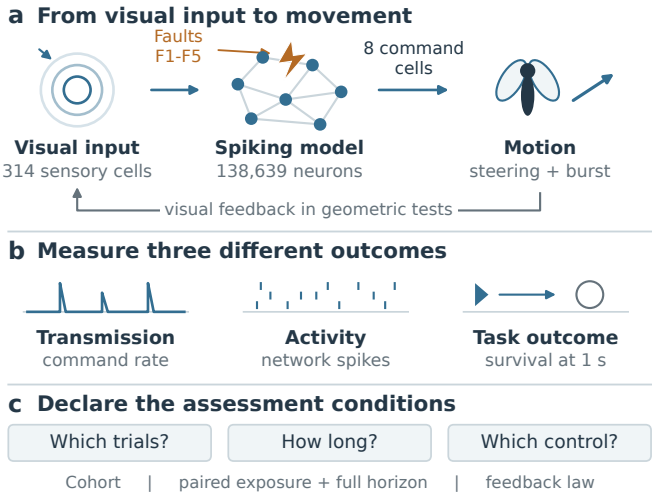



Fig. 1. Assessment overview. (a) Model flow with visual feedback in geometric tests. (b) Distinct transmission, activity and bounded task endpoints. (c) Required assessment conditions. Icons and traces are schematic; Supplementary Fig. S8 retains mapped wiring.

## II. Related Work

Connectome models connect anatomy to executable sensorimotor hypotheses [3], while graph statistics characterize wiring rather than dynamical reliability [2]. A recent sparsification study investigates compression across connection strength, placement, state and task, including active-state overexcitation [4]. Endpoint-dependent damage responses have also been reported on the male central nervous system connectome [5]. Those studies investigate how damage or sparsification changes function. Our assessment comparison explicitly separates paired exposure, full observation horizons and neural versus task endpoints. It introduces no perturbation operator or claim of superior fault tolerance. Bilateral integration measured in the giant-fiber circuit [6] cautions against treating the controller's directional failure as established fly behavior.

SpikeFI [7], SpikingJET [8] and enpheeph [9] provide SNN fault-injection frameworks. Population-coded regression

[1]Review package: `assessment-v8`, including `supplement.pdf` and reproducible paired analyses. Repository: https://github.com/Scophield/drosophila-fault-injection; this local revision is not yet published there.

[10] and scientific computing [11] assess application outputs. Loihi 2 deployment [12] establishes implementation feasibility. NeuroBench [13] and embodied benchmarking [14] motivate separate functional/resource measures. We examine assessment definitions rather than benchmark injector implementations.

## III. Model, Perturbations and Endpoints

Evidence is tiered: prospective geometry-corrected experiments provide the principal task evidence; archive reanalysis supplies cohort/exposure sensitivity; reconstructed historical selectors are diagnostic only. Missing historical metadata are not inferred from current code.

### A. Spiking Substrate and Readout

The model uses the released signed connectivity and parameters [3], implemented with Brian 2 [15]. There are 138,639 neurons and 15,091,983 connection-table rows representing 54.5 million contacts. Between events, each unit follows

$$\frac{dv_i}{dt} = \frac{v_0 - v_i + g_i}{20\,\mathrm{ms}}, \qquad \frac{dg_i}{dt} = -\frac{g_i}{5\,\mathrm{ms}}. \tag{1}$$

Here $g_i$ is voltage-valued synaptic drive. Events add signed contact counts times $0.275\,\mathrm{mV}$ after 1.8 ms. Spikes above $-45\,\mathrm{mV}$ reset voltage to $v_0 = -52\,\mathrm{mV}$ and drive to zero. Both equations pause during 2.2 ms refractoriness, retained in sensory cells unlike upstream stimulation. This variant is tested below. Linear integration advances the dynamics; stored initial states are restored before each trial. The time step used in the archived simulations was not documented; fresh runs pin 0.1 ms.

Independent 150 Hz Poisson trains drive 104 LC4 and 210 LPLC2 cells (Fig. 1). Each external event adds 68.75 mV to the target voltage. Other cells receive no background drive. LC4/LPLC2 are associated with looming processing, including measured LPLC2 motion opponency [16]; the demo's receptive fields are assumptions, not measured maps. The command readout is eight cells of types DNp01, DNa02, DNp04 and DNb01 [17]. For duration $T = 0.6\,\mathrm{s}$, command rate is $r = N_{\mathrm{cmd}}/(8T)$. Latency is the first spike in this group, not a command decision time. Total activity $S$ counts all model spikes.

### B. Functional Perturbations

F1 multiplies each weight by $\exp(\sigma Z - \sigma^2/2)$, $Z \sim \mathcal{N}(0,1)$, with $\sigma = 0.05, 0.10, 0.20, 0.50$. This preserves expected weight; it does not implement bit flips. F2 independently zeroes weighted connection-table rows with probabilities $f = 0.05, 0.10, 0.25, 0.50$, not individual contacts; expected lost contact mass is $f$. F3 silences all outgoing connections of a random fraction $0.01, 0.02, 0.05, 0.10$ of eligible units, protecting sensory and command cells. Those units still receive input and spike. F4 targets the top 1%, 2% or 5% by outgoing contact count; its archived selections lack cell manifests and have protection inconsistencies. F5 silences LC4, LPLC2 or both output channels.

There are 12 fault-seeded 600 ms trials per condition. F1–F3 draw realizations from SHA-256 seeds over (phase, condition, trial). Archived open-loop stimulus streams advance without explicit per-trial reseeding; dispersion combines fault and stimulus variation. Fresh code saves selected cell identities, input hashes and backend metadata. Corrected selection excludes protected FlyWire identities and breaks ties by identity. It has not generated the historical F4 data.

### C. Scoring and Controls

Transmission requires $r \geq \theta = 51.4\,\mathrm{Hz}$, rounding $(94.5833 + 8.2292)/2 = 51.40625\,\mathrm{Hz}$ from separate ten-trial sensory and random-drive calibrations. The operational joint rule is

$$r \geq 51.4\,\mathrm{Hz}, \qquad r \leq c\,r_{\mathrm{H}}, \qquad S \leq c\,S_{\mathrm{H}}, \tag{2}$$

with healthy medians $r_{\mathrm{H}} = 94.6875\,\mathrm{Hz}$, $S_{\mathrm{H}} = 39{,}694$ and $c = 2$ (sensitivity: 1.5, 3, 4). This operational sensitivity criterion is not a hardware-, power- or safety-validated failure threshold. Joint passes are nested in transmission; disagreement does not establish predictive accuracy. The 288 records mix 216 main-sweep faults and 72 HUB/EIm/random-inhibitory controls, not 288 independent population samples. Twelve healthy references provide resubstitution checks only. Median intervals use 20,000 trial resamples, seed 0; relative effects independently resample fault and healthy groups. Dispersion bars instead show 10th–90th percentiles.

The target-permuted null preserves sources, weights, signs and incoming row counts while changing convergence and weighted input. It fails intact transmission, so supplies a specificity control rather than a functioning fault-tolerance alternative. Supplementary Table S4 reports realized control damage. Expected contact-mass matching does not imply realized edge equality.

### D. Stopping Times and Observation Horizons

Legacy 20 ms frames couple a readout-driven planar agent to a pursuing threat. Collision is recorded at 0.05 m with a nominal 1 s horizon. Six calibration episodes overlap the healthy evaluation; all fault episodes share calibration seeds. We treat these records as illustrative. For paired cumulative spikes $S_{\mathrm{F}}, S_{\mathrm{H}}$ and recorded durations $T_{\mathrm{F}}, T_{\mathrm{H}}$, define

$$R_{\mathrm{whole}} = \frac{S_{\mathrm{F}}(T_{\mathrm{F}})}{S_{\mathrm{H}}(T_{\mathrm{H}})}, \quad R_{\mathrm{common}} = \frac{S_{\mathrm{F}}(T_*)}{S_{\mathrm{H}}(T_*)}, \quad R(t) = \frac{S_{\mathrm{F}}(t)}{S_{\mathrm{H}}(t)}, \tag{3}$$

where $T_* = \min(T_{\mathrm{F}}, T_{\mathrm{H}})$ and $t \leq T_*$. Matching unequal stopping times differs from selecting an earlier horizon. All selected 200–500 ms windows precede included terminations; full-hub pairs reach 1000 ms. Diagnostics retain every pair. The $3^3 = 27$ ordered draws with replacement of three seed clusters are bootstrap resamples, not independent experiments or confirmatory coverage.

### E. Prospective Geometric Validation

The prospective protocol uses body-frame bearing, actual-separation expansion and within-frame collision checks. Nine passive 400 ms episodes ($-60/-30/0°$, seeds 0–2) fix calibration before testing. At frame $k$, let $d_k$ be summed left-minus-right command rates and $a_k$ their eight-cell mean. Bars denote means over the latest five frames (all available frames initially). Steering is

$$\omega_k = -400\,\mathrm{deg/s}\ \mathrm{clip}\left(\frac{\bar{d}_k - b}{\ell}, -1, 1\right), \tag{4}$$

with frozen $b = -20$ Hz, $\ell = 70$ Hz and positive data sign. Body-frame $+30°$ is left, $-30°$ right; positive heading turns left. For example, $\bar{d} = 50$ Hz gives $\omega = -400°$/s (right), whereas $\bar{d} = -90$ Hz gives $+400°$/s (left). These define the implemented sign convention, not a biological response. Triggering requires current $\bar{a}_k$ and preceding unsmoothed $a_{k-1}$ both to reach 18.125 Hz. The first qualifying frame schedules one burst 250 ms after its end. Burst speed is 0.35 m/s for 450 ms with fixed heading; otherwise speed is 0.05 m/s and heading is capped at $\pm 100°$. Cue onset is 100 ms; original steering has no onset gate.

The first 48 tests use seeds 6–11, $\pm 30°$, a 1 s horizon, healthy/F2_25/F2_50 and readout disabling. Healthy controls on seeds 12–17 compare the original law, pre-cue gating ($\omega = 0$ before 100 ms), offset only ($\bar{d} = 0$) and no steering ($\omega = 0$), retaining the neural trigger. The new factorial comparison reuses 24 byte-identical original/gated healthy records and adds 48 F2_25/F2_50 episodes on these same seeds and bearings. Calibration, neural-source snapshots, input hashes and package versions are verified unchanged. Brian2 seed $20000 + s$ and deletion seed $10000 + s$ are reset per episode. Masks are shared across controllers/bearings and nested across doses. Healthy outcomes were known before this extension; no controller is retuned.

For matched binary survival $Y_H, Y_F$, paired loss is $L = I(Y_H = 1, Y_F = 0)$, gain is $G = I(Y_H = 0, Y_F = 1)$ and net loss $D = Y_H - Y_F = L - G$. The interaction is $D_{gate} - D_{original}$, averaging both bearings within seed. Exploratory conditional ranges use 20,000 six-cluster resamples, seed 0; 72 records are not independent replicates. Activity compares each faulty record to the same-law healthy reference at 200–500 ms and common duration. The NumPy backend uses Brian2 2.10.1, NumPy 2.5.4 and 0.1 ms steps on a Xeon W-2223. Supplementary Tables S17–S22 give paired records, activity, thresholds and compute; protocol v8 gives reproduction commands.

## IV. Results

### A. Held-Out Whole-Connectome Tests

Frozen calibration yielded healthy collision-free survival in 5/6 cases at $-30°$ and 0/6 at $+30°$ (Fig. 2). The corresponding counts were 4/6 and 0/6 under 25% deletion; 50% deletion and readout disabling yielded 0/12 each. On paired negative-bearing seeds, deletion lost one and five healthy

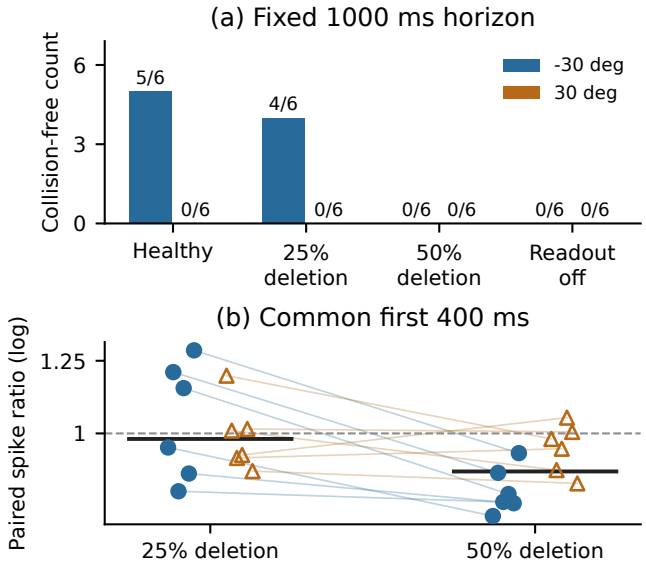



Fig. 2. First prospective campaign, seeds 6–11. (a) Collision-free counts at 1000 ms, six seeds per bearing. (b) Every 400 ms activity pair: circles $-30°$, triangles $+30°$, black median. Thin lines connect the same seed/bearing across deletion doses. Shared seeds are not independent replicates. This activity limit did not distinguish the observed collisions; no predictive detector was fitted.

TABLE I
FROZEN CONTROLLER × FAULT COMPARISON, SEEDS 12–17. SIX CLUSTERS SHARE BOTH BEARINGS AND ALL CONDITIONS. L/G: PAIRED HEALTHY-TO-FAULT LOSSES/GAINS OUT OF 12; HEALTHY SURVIVORS AT RISK ARE FOUR/SIX UNDER ORIGINAL/GATED LAWS. HEALTHY ROWS REUSE PRESERVED RECORDS.

| Law | Network | $-30°$ | $+30°$ | Total | L/G |
|---|---|---|---|---|---|
| Original | Healthy | 4/6 | 0/6 | 4/12 | — |
| Original | 25% deletion | 4/6 | 0/6 | 4/12 | 1/1 |
| Original | 50% deletion | 0/6 | 0/6 | 0/12 | 4/0 |
| Pre-cue gate | Healthy | 6/6 | 0/6 | 6/12 | — |
| Pre-cue gate | 25% deletion | 3/6 | 0/6 | 3/12 | 3/0 |
| Pre-cue gate | 50% deletion | 0/6 | 0/6 | 0/12 | 6/0 |

survivors, respectively; the positive bearing was already at floor. These are six-cluster conditional observations. For 50% deletion, matched spike ratios were 0.748/0.845/0.890/1.012 over 200/300/400/500 ms. Neither deletion condition exceeded twice its paired healthy activity in those windows, despite eight and twelve collisions. Thus this activity limit did not flag these outcomes. It is distinct from the open-loop joint rule.

### B. Controller-by-Fault Comparison

All 12 first-campaign healthy tests turned $-11.429°$ before the 100 ms cue while recording zero spikes. Equation (4) produces $-114.286°$/s at zero readout. Readout disabling removes both this offset and the neural trigger, so its comparison alone cannot isolate stimulus-dependent steering.

The 72-record matrix directly pairs each fault with its same-law healthy reference (Table I). The healthy gate advantage is two episodes out of 12 (+16.7 percentage points); its six-cluster conditional range includes zero. Offset-only and no-steering healthy controls triggered bursts in 11/12 and 12/12

cases but yielded 0/12 survival (Supplementary Tables S12–S14). These compare complete feedback laws, which can change sensory trajectories and trigger timing.

At 25% deletion, gated-minus-original net task loss was +25.0 percentage points (conditional 95% range 8.3 to 41.7). At 50% deletion, gated-minus-original net task loss was +16.7 percentage points (conditional 95% range 0.0 to 33.3). The 25% range excludes zero, but only 3 seed clusters have nonzero interactions; a post-hoc symmetric-null sign-flip sensitivity gives $p = 0.25$ (Supplementary Section 13). At 50% deletion, both controllers yield 0/12; their net-loss contrast reflects different healthy baselines. Loss and gain counts are retained separately; a healthy success difference alone is not a fault-tolerance comparison.

Among 48 faulty episodes, 41 collided; none of these collisions was flagged at any tested $c = 1.5, 2, 3, 4$ over 200–500 ms. At 500 ms, $c = 1.5$ instead flagged 2 surviving fault episodes. This activity-only criterion did not distinguish the observed collisions under these settings; it does not establish that spike count is generally uninformative. Supplementary Table S22 and the threshold CSV report the full sensitivity.

### C. Transmission Under Five Perturbation Families

Healthy command rate is 94.69 Hz (95% median CI 94.06–95.42), with first-spike latency 4.05 ms. Equal-sized random drive yields approximately 8.2 Hz despite greater activity; the intact target-permuted graph yields zero. This specificity does not isolate a fault-tolerance mechanism.

The supplement supplies every main dose, recorded control damage and selected-dose intervals. F1 at $\sigma = 0.50$ retained 99.4% of median rate, with 10th–90th percentiles 90.40–100.81 Hz versus healthy 93.96–95.63 Hz. F3 at 10% retained 100.4%. These sampled medians do not establish equivalence.

F2 retained 97.0%, 98.3%, 89.2% and 76.5% of median command rate at 5%, 10%, 25% and 50% deletion. The 95% bootstrap interval for the ratio of median rates at 50% was 75.1–79.0%. Latency rose from 4.05 to 5.40 ms (ratio 1.33, interval 1.25–1.40). All sampled deletion trials passed transmission; a separate 75 Hz control passed only 10/12 at 50% deletion. F5 retained 79.8% or 67.9% when removing LC4 or LPLC2 outputs; silencing both eliminated transmission. Supplementary plots show all doses and dispersion.

Twelve new 600 ms runs tested the sensory refractory variant on three paired seeds. Median paired 50%-deletion retention was 76.3% at 2.2 ms versus 79.2% at zero sensory refractory time. Healthy median rates were 94.8 and 103.1 Hz. These conditional observations do not establish variant invariance.

### D. Historical Hub Records and Activity Scoring

The historical 1% F4 condition yielded 119.06 Hz and 49,272.5 median spikes (1.24× healthy). Cell manifests are missing; reconstructed selection includes protected readout cells, and implementations differ. Historical contrasts cannot be attributed to cell sign alone. The artifact's `audit/verified-hubs` contains selector reconstructions, not recovered historical manifests; Supplementary Table S4 gives recorded damage.

TABLE II
POST-HOC COHORT SENSITIVITY, $c = 2$. T: TRANSMISSION; J: JOINT PASSES. EIM: COMPOSITION-MATCHED CONTROLS. ROWS SUCCESSIVELY ISOLATE THE INHIBITORY SUBSET, ALL TARGETED-HUB COHORTS, THEN THE MAIN NON-HUB SWEEP BY REMOVING WHOLE CONDITIONS. REMAINING DISAGREEMENTS ARE RANDOM-MASS TRIALS 3 AND 9; MIXTURES DO NOT ESTIMATE POPULATION FAILURE PROBABILITY.

| Included records | $n$ | T | J | T−J |
|---|---|---|---|---|
| All faults/controls | 288 | 276 | 262 | 14 |
| Exclude inhibitory hubs | 276 | 264 | 262 | 2 |
| Exclude F4/HUB/EIm | 204 | 192 | 190 | 2 |
| Main F1/F2/F3/F5 | 180 | 168 | 168 | 0 |

The inhibitory subset had 847,476.5 median spikes (21.35× healthy), failing the activity criterion in 12/12 trials. The random inhibitory expected-contact-mass control had median 50,323 spikes, with two extreme trials (327,246 and 271,058). Individual records and realized damage are supplied.

Across 288 fault/control records, 276 passed transmission and 262 jointly. Twelve extra rejections arose from inhibitory hubs and two from random-mass records (Table II). Excluding targeted hubs left two disagreements among 204 records; the main non-hub sweep had none. Counts were unchanged for $c = 1.5, 2, 3, 4$. All 12 healthy reference trials passed, but resubstitution does not estimate held-out false alarms or calibrate the limit.

### E. Legacy Evaluation and Stopping-Time Matching

Legacy looming used elapsed time and heading canceled from receptive-field bearing. Six threshold-calibration logs duplicate healthy evaluation records; all fault episodes share calibration seeds. Four-bearing/three-seed survival was 10/12 for 25% deletion, 0/12 for 50%, 12/12 for full hubs and 6/12 for inhibitory hubs. Eight healthy records stopped during a burst. These are compromised recorded-horizon illustrations, not corrected geometric tests (Supplementary Table S5).

For legacy 50% deletion, early collisions shorten fault records relative to healthy references. Matching each pair to $T_*$ changes the median spike ratio from $R_{\mathrm{whole}} = 0.382$ to $R_{\mathrm{common}} = 0.729$. Fixed 200–500 ms ratios span 0.723–0.745. This stopping-time mismatch is distinct from the horizon effect below; neither ratio defines an activity budget.

### F. Late Activity Under Equal Durations

All full-hub and paired healthy records last 1000 ms, so $R_{\mathrm{whole}} = R_{\mathrm{common}}$ for each pair. No ratio exceeds twofold through 800 ms. Six first crossings occur at 840–900 ms, all at $-60°$ or $-30°$ on calibration seeds (Fig. 3). Thus selecting 400 ms excludes late exceedances; it does not remove unequal exposure. Pooled medians rise from 1.362 at 400 ms to 1.703 at 1000 ms. Inhibitory hubs exceed twofold in every pair at all 200–500 ms windows. Individual trajectories and three-seed sensitivity are supplied; sparse conditional bootstrap ranges do not establish independent reliability.

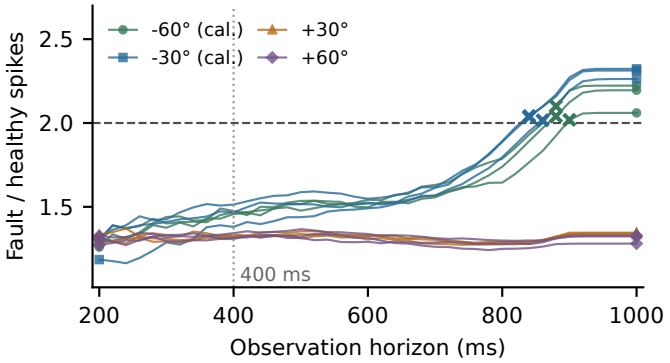



Fig. 3. Legacy full-hub cumulative activity ratios. Both members of every pair last 1000 ms. Marker/color encode bearing; three same-bearing traces are shared seeds. Negative bearings are calibration scenarios. Crosses mark first twofold crossings at 840–900 ms. The first 400 ms excludes late activity rather than correcting unequal durations.

The privileged geometry rule survives both bearings at 800 and 1000 ms but collides by 1200 ms. This establishes bounded task feasibility with identical actuator limits, without a neural resource-cost comparison. The supplement records each seed, termination and generating manifest.

## V. Implications, Limitations and Conclusion

Fault comparisons require declared cohorts, criteria and stopping rules. Targeted-hub exclusion leaves two disagreements; the main non-hub sweep shows none. Duration matching changes deletion ratios, whereas early hub horizons omit late activity. The six recorded one-second exceedances remain after duration matching. Resource-failure prediction requires independent budgets and held-out healthy false-alarm measurements.

One connectome, selected drives and small samples limit transfer. The intact null is nonfunctional; missing hub manifests and compromised legacy evaluation limit interpretation. Post-hoc exclusions and windows require prospective confirmation. Software perturbations support no hardware protection policy without mapped resources and faults.

The new factorial comparison directly pairs faults within each frozen control law. Gating improves healthy counts from 4/12 to 6/12, yet at 25% deletion yields 3/12 versus 4/12. The original law's zero net loss at this dose contains one loss and one gain, not absence of fault effects. The positive conditional interaction range rests on three nonzero seed clusters; post-hoc sign-flip sensitivity does not supply confirmatory evidence. At 50% deletion both laws fail all 12 episodes, so their net-loss difference reflects healthy baseline success. Six clusters, directional floors, negative/frontal calibration coverage and assumed receptive fields limit generalization. Some survivors are still bursting at 1 s; bounded survival does not establish eventual escape.

Report transmission, activity and bounded task outcomes with cohort denominators, paired exposure, analysis horizon, termination reason and direction-specific results. Retain paired losses and gains as well as net effects. A pass fraction cannot certify reliability or a safety budget.

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